

## lab09

May 7, 2026

### 0.1 Exercise

For the transfer function  $H(z) = \frac{1}{z-0.5}$ , determine whether the system is stable. Justify your answer.

#### 0.1.1 Solution 1

$$H(z) = \frac{z^{-1}}{1 - 0.5z^{-1}}.$$

The pole is at

$$z = 0.5.$$

The algebraic expression alone does not determine the impulse response, because the ROC is not specified.

If the system is causal, then

$$\text{ROC} : |z| > 0.5.$$

Then

$$H(z) = z^{-1} \frac{1}{1 - 0.5z^{-1}} = z^{-1} \sum_{n=0}^{\infty} (0.5)^n z^{-n}$$

so

$$H(z) = \sum_{n=0}^{\infty} (0.5)^n z^{-(n+1)}.$$

Therefore,

$$h[n] = (0.5)^{n-1} u[n-1] = \{0, 1, 0.5, 0.25, \dots, (0.5)^{n-1}, \dots\}.$$

Now test absolute summability:

$$\sum_{n=-\infty}^{\infty} |h[n]| = \sum_{n=1}^{\infty} (0.5)^{n-1} = 1 + 0.5 + 0.25 + \dots = 2 < \infty.$$

Thus, if the system is causal, it is stable.

However, if the ROC is

$$|z| < 0.5,$$

then

$$H(z) = \frac{1}{z - 0.5} = -\frac{2}{1 - 2z} = -2 \sum_{n=0}^{\infty} (2z)^n$$

so

$$H(z) = - \sum_{n=0}^{\infty} 2^{n+1} z^n.$$

Since

$$z^n = z^{-(-n)},$$

this corresponds to

$$h[-n] = -2^{n+1}, \quad n \geq 0.$$

Equivalently,

$$h[n] = -2^{1-n} u[-n].$$

This is not absolutely summable:

$$\sum_{n=-\infty}^0 |h[n]| = \infty.$$

Therefore,

$$\text{Stable for ROC } |z| > 0.5, \quad \text{unstable for ROC } |z| < 0.5.$$

### 0.1.2 Remark

There is another way to get the impulse response. We know from theory that the most general form of a causal LTI is:

$$\sum_{l=0}^m b_l \cdot y[n-l] = \sum_{l=0}^k a_l \cdot x[n-l]$$

which at the level of the  $z$ -transforms

$$B(z) \cdot Y(z) = A(z) \cdot X(z) \quad \rightarrow \quad Y(z) = \frac{A(z)}{B(z)} X(z) \quad \rightarrow \quad H(z) = \frac{A(z)}{B(z)}.$$

In our case  $A(z) = z^{-1}$ ,  $B(z) = 1 - 0.5z^{-1}$  whence  $a[n] = \delta[n-1] = \{0, 1\}$  and  $b[n] = \delta[n] - 0.5\delta[n-1] = \{1, -0.5\}$ .

Therefore the system is defined as  $y[n] - 0.5y[n-1] = x[n-1]$  or  $y[n] = 0.5y[n-1] + x[n-1]$ .

For the impulse response:  $x[n] = \delta[n]$ ,  $y[n] = h[n]$ .

Thus,  $h[n] = 0.5h[n-1] + \delta[n-1]$ .

That is:

$$h[0] = 0.5 \cdot 0 + 0 = 0$$

$$h[1] = 0.5 \cdot 0 + 1 = 1$$

$$h[2] = 0.5 \cdot 1 + 0 = 0.5$$

$$h[3] = 0.5 \cdot 0.5 + 0 = 0.25$$

$$h[n] = 0.5^{n-1}$$

### 0.1.3 Solution 2

Consider the input

$$x[n] = e^{i\omega n}.$$

(the sampling period  $T$  is not specified therefore we work in time units of  $T$  (equivalently assume that  $T = 1$ ) therefore  $\omega T \rightarrow \omega \in [-\pi, \pi)$ ).

For an LTI system,

$$y[n] = H(e^{i\omega}) e^{i\omega n}.$$

Evaluate the transfer function on the unit circle:

$$H(e^{i\omega}) = \frac{1}{e^{i\omega} - 0.5}.$$

Its magnitude is

$$|H(e^{i\omega})| = \frac{1}{|e^{i\omega} - 0.5|}.$$

Since  $e^{i\omega}$  lies on the unit circle, the minimum distance to 0.5 is

$$|1 - 0.5| = 0.5,$$

so

$$|H(e^{i\omega})| \leq 2.$$

Thus every sinusoidal input is scaled by a finite factor:

$$|y[n]| \leq 2|x[n]|.$$

This argument is valid only if the unit circle belongs to the ROC.

For the causal ROC

$|z| > 0.5,$

the unit circle is included, so the system is stable.

## 0.2 Exercise

Study the stability of the system with transfer function

$$H(z) = \frac{1}{z^2 - z + 0.5}$$

and determine its impulse response.

### 0.2.1 Solution

The two poles are  $p_{1,2} = \frac{1 \pm i}{2}$   $|p_{1,2}| < 1 \rightarrow$  stable causal system (the unit circle is within the ROC which is  $|z| > \max(|p_1|, |p_2|)$ )

|   |     |                   |                     |                     |                           |                       |                        |     |  |                                                      |  |
|---|-----|-------------------|---------------------|---------------------|---------------------------|-----------------------|------------------------|-----|--|------------------------------------------------------|--|
|   |     |                   |                     |                     | $\frac{1}{z^2 - z + 0.5}$ |                       |                        |     |  |                                                      |  |
| 1 | +0z | +0z <sup>2</sup>  | +0z <sup>3</sup>    | +0z <sup>4</sup>    | +0z <sup>5</sup>          | +0z <sup>6</sup>      | +0z <sup>7</sup>       | ... |  | $z^2 - z + 0.5$                                      |  |
| 1 | -z  | +0.5              |                     |                     |                           |                       |                        |     |  | $z^{-2} + z^{-3} + 0.5z^{-4} + 0z^{-5} - 0.25z^{-6}$ |  |
|   | z   | -0.5              |                     |                     |                           |                       |                        |     |  |                                                      |  |
|   | z   | -z <sup>2</sup>   | +0.5z               |                     |                           |                       |                        |     |  |                                                      |  |
|   |     | 0.5z <sup>2</sup> | -0.5z               |                     |                           |                       |                        |     |  |                                                      |  |
|   |     | 0.5z <sup>2</sup> | -0.5z <sup>3</sup>  | +0.25z <sup>2</sup> |                           |                       |                        |     |  |                                                      |  |
|   |     |                   | -0.25z <sup>2</sup> |                     |                           |                       |                        |     |  |                                                      |  |
|   |     |                   | -0.25z <sup>2</sup> | +0.25z              | -0.125                    |                       |                        |     |  |                                                      |  |
|   |     |                   |                     | -0.25z              | +0.125                    |                       |                        |     |  |                                                      |  |
|   |     |                   |                     | -0.25z              | +0.25z <sup>-1</sup>      | -0.125z <sup>-2</sup> |                        |     |  |                                                      |  |
|   |     |                   |                     |                     | -0.125                    | +0.125z <sup>-1</sup> |                        |     |  |                                                      |  |
|   |     |                   |                     |                     | -0.125                    | +0.125z <sup>-1</sup> | -0.0625z <sup>-2</sup> |     |  |                                                      |  |
|   |     |                   |                     |                     |                           | 0.0625z <sup>-2</sup> |                        |     |  |                                                      |  |

So  $H(z) = z^{-2} + z^{-3} + 0.5z^{-4} + 0z^{-5} - 0.25z^{-6} - 0.25z^{-7} - 0.125z^{-8} + 0z^{-9} + \dots$

That is  $h[n] = \{0, 0, 1, 1, 0.5, 0, -0.25, -0.25, -0.125, 0, \dots\}$

### 0.2.2 Alternative solution

$$H(z) = \frac{z^{-2}}{1 - z^{-1} + 0.5z^{-2}}$$

$$A(z) = z^{-2}, B(z) = 1 - z^{-1} + 0.5z^{-2} \rightarrow a[n] = \{0, 0, 1\}, b[n] = \{1, -1, 0.5\}$$

$$y[n] - y[n-1] + 0.5y[n-2] = x[n-2] \rightarrow y[n] = y[n-1] - 0.5y[n-2] + x[n-2]$$

$$\rightarrow h[n] = h[n-1] - 0.5h[n-2] + \delta[n-2]$$

$$h[0] = 0 - 0.5 \cdot 0 + 0 = 0$$

$$h[1] = 0 - 0.5 \cdot 0 + 0 = 0$$

$$h[2] = 0 - 0.5 \cdot 0 + 1 = 1$$

$$h[3] = 1 - 0.5 \cdot 0 + 0 = 1$$

$$h[4] = 1 - 0.5 \cdot 1 + 0 = 0.5$$

$$h[5] = 0.5 - 0.5 \cdot 1 + 0 = 0$$

$$h[6] = 0 - 0.5 \cdot 0.5 + 0 = -0.25$$

$$h[7] = -0.25 - 0.5 \cdot 0 + 0 = -0.25$$

$$h[8] = -0.25 - 0.5 \cdot (-0.25) + 0 = -0.125$$

$$h[9] = -0.125 - 0.5 \cdot (-0.25) + 0 = 0$$

So

$$h[n] = \{0, 0, 1, 1, 0.5, 0, -0.25, -0.25, -0.125, 0, \dots\}$$

### 0.2.3 Yet another solution

$$p = p_1 = p_2^* = 1/\sqrt{2}e^{i\pi/4}$$

$$H(z) = \frac{1}{i} \left( \frac{1}{z-p} - \frac{1}{z-p^*} \right) = H_1(z) - H_2(z)$$

$$\text{From a previous exercise } h_1[n] = (1/i)p^{n-1}u[n-1], \quad h_2[n] = (1/i)p^{*(n-1)}u[n-1]$$

$$\begin{aligned} h[n] &= h_1[n] - h_2[n] = 2\text{Im}(p^{n-1}u[n-1]) = 2 \left[ \frac{\sin((n-1)\pi/4)}{(\sqrt{2})^{n-1}} \right] u[n-1] = \\ &= \{0, 0, 1, 1, 0.5, 0, -0.25, -0.25, -0.125, 0, \dots\} \end{aligned}$$

[ ]:

[ ]: